LA HW

Included exercises: 1-2, 9, 12, 14-15, 37, 44

Exercises 1-8

In Exercises 1–8, compute the determinant of each matrix.

Original Exercise 1

$$\begin{bmatrix} 6 & 2 \\ -3 & -1 \end{bmatrix}$$

Original Exercise 2

$$\begin{bmatrix} 4 & 5 \\ 3 & -7 \end{bmatrix}$$

Original Exercise 9

the $(1, 2)$-cofactor

Original Exercise 12

the $(3, 3)$-cofactor

Exercises 13-20

In Exercises 13–20, compute the determinant of each matrix A by a cofactor expansion along the indicated row.

$$\begin{bmatrix} 2 & -1 & 3 \end{bmatrix} \qquad \begin{bmatrix} 1 & -2 & 2 \end{bmatrix}$$

Original Exercise 14

$$\begin{bmatrix} 1 & -2 & 2 \\ 2 & -1 & 3 \\ 0 & 1 & -1 \end{bmatrix}$$
along the second row

Original Exercise 15

$$\begin{bmatrix} 1 & -2 & 2 \\ 2 & -1 & 3 \\ 0 & 1 & -1 \end{bmatrix}$$
along the third row

Exercises 37-44

In Exercises 37–44, find each value of c for which the matrix is not invertible.

Original Exercise 37

$$\begin{bmatrix} 3 & 6 \\ c & 4 \end{bmatrix}$$

Original Exercise 44

$$\begin{bmatrix} c & 9 \\ 4 & c \end{bmatrix}$$